

Interpretable Quality Assessment for Machine Translation: A Study of Prompting Strategies, Entity-Aware Methods and Semantic Inference

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Abstract—Evaluating machine translation quality in specialized domains remains challenging, particularly for monolingual users who lack reference translations. This work extends the AskQE framework—a question-answering pipeline for semantic quality estimation—by integrating domain-specific Named Entity Recognition (NER), advanced Natural Language Inference (NLI) metrics, and systematic prompt engineering. Using Qwen2.5-3B-Instruct on BioMQM, a biomedical translation dataset spanning five language pairs with MQM severity annotations, we demonstrate enhanced discriminative power: the NER-guided extension achieves a more gradual score reduction between Neutral and Critical errors, indicating improved sensitivity to clinically significant translation failures. Our dual-metric framework combining BART-based NLI classification and LLM-as-Judge achieves 69.77% inter-model agreement, but only the NLI classification task leads to improvements in performance. Prompt ablation studies show that Few-Shot strategies (P1) and Chain-of-Thought strategies (P2) yield consistent improvements across severity levels. These findings demonstrate that targeted architectural refinements can significantly enhance both sensitivity and interpretability of monolingual quality estimation in high-stakes translation scenarios. The code and implementation are available at https://github.com/Simone280802/AskQE_DNLP_2025-2026.

I. PROBLEM STATEMENT

Evaluating Machine Translation (MT) is a critical challenge, especially for monolingual users in high-stakes domains like healthcare. Traditional evaluation approaches lack both the semantic depth to capture meaning-altering errors and the interpretability needed to help users identify potentially consequential mistakes [9].

To enhance interpretability, the AskQE framework [4] estimates quality by analyzing semantic consistency through a question-answering (QA) pipeline. Formally, given a source sentence s , a set of questions $\mathcal{Q} = \{q_1, \dots, q_k\}$ is generated. These questions are answered using both s and the backtranslation s_{bt} , yielding answer sets $\mathcal{A}^{(s)}$ and $\mathcal{A}^{(bt)}$. Translation quality is quantified by the semantic divergence between these sets.

While ASKQE provides a solid foundation for monolingual quality estimation, there remain opportunities to enhance its performance in specialized contexts, particularly regarding its reliance on general prompting and string based similarity. To

refine this approach, this study integrates Named Entity Recognition (NER) to guide the Question Generation (QG) process, ensuring the model emphasizes critical biomedical entities such as drug names and dosages. This refinement increases the system’s sensitivity to factual nuances within domain specific translations. Furthermore, to move beyond the constraints of lexical overlap, Natural Language Inference (NLI) and LLM-as-a-Judge frameworks are utilized to evaluate logical entailment between source and backtranslated responses. By comparing the results of the two models the proposed methodology achieves a more robust assessment of translation fidelity. Finally, the Question Answering (QA) phase is optimized through the systematic evaluation of Few-Shot, Chain-of-Thought (CoT) and Concise Constraint prompting strategies.

II. METHODOLOGY

A. Baseline: Vanilla AskQE with Qwen2.5-3B

We replicate the AskQE pipeline by replacing the original LLaMA-3 70B backbone with Qwen2.5-3B-Instruct [12], a substantially smaller model (3B vs. 70B parameters). This substitution is motivated by computational constraints and serves as a controlled baseline for our subsequent extensions: all configurations share the same underlying LLM.

B. Metrics Extension

Beyond lexical overlap metrics (F1, Exact Match, BLEU, chrF) and embedding-based similarity (SBERT), we propose evaluating the semantic consistency between source and backtranslated answers through Natural Language Inference (NLI).

We model the semantic relationship between these answers as a classification problem mapping to a label space $L = \{\text{ENTAILMENT}, \text{NEUTRAL}, \text{CONTRADICTION}\}$. The core hypothesis is that a high-quality translation should preserve the semantic content of the source, leading to entailment between a_s and a_{bt} . Conversely, translation failures manifest as contradictions (critical semantic conflicts).

We implement two complementary approaches to perform this classification:

1) *NLI Classifier*: We employ BART-large [5] fine-tuned on the MultiNLI corpus, a standard model for NLI tasks. Given the concatenated input $[a_s; a_{bt}]$, the model outputs logits $\mathbf{z} \in \mathbb{R}^3$. These are converted to a probability distribution via the softmax function:

$$P_{\text{NLI}}(y = c \mid a_s, a_{bt}) = \frac{\exp(z_c)}{\sum_{j \in L} \exp(z_j)} \quad (1)$$

where $c \in L$ and z_c is the logit corresponding to class c . The predicted label is:

$$\hat{y}_{\text{NLI}} = \arg \max_{c \in L} P_{\text{NLI}}(y = c \mid a_s, a_{bt}) \quad (2)$$

This model serves as our reference classifier, as it was explicitly trained for NLI tasks.

2) *LLM-as-Judge*: Alternatively, drawing on recent work utilizing LLMs as evaluators [14, 10], we employ a generative LLM (Qwen-2.5 3B Instruct) to estimate classification probabilities via the *first-token probability* method [3].

The possible labels the LLM can assign to the answers are $L = \{\text{Entailment}, \text{Neutral}, \text{Contradiction}\}$.

Denote the text output of LLMs as a sequence of tokens $\mathbf{w} = [w_1, w_2, \dots, w_k]$, with $w_i \in \mathcal{V}$, where k is the length of the generated sequence and $\mathcal{V}$ is the vocabulary used by the LLM.

We extract only the logits of the first token w_1 corresponding to our option set, denoted as $\mathbf{s}_{w_1} = [s_j]_{j \in L}$. To transform $\mathbf{s}_{w_1}$ into a probability distribution $\mathbf{p}$, we utilize the softmax function with a temperature parameter τ :

$$p(j) = \frac{\exp(s_j/\tau)}{\sum_{k \in L} \exp(s_k/\tau)}, \quad \forall j \in L \quad (3)$$

The predicted label corresponds to the class with the highest probability in $\mathbf{p}$:

$$\hat{y} = \arg \max_{c \in L} p(c) \quad (4)$$

We also track the confidence of each prediction as the probability assigned to the predicted label:

$$\text{conf}(\hat{y}) = \max_{c \in L} p(c) \quad (5)$$

This allows us to analyze not only *what* label is predicted, but *how certain* the model is about its prediction.

C. NER extension

We introduce a verticalized extension that integrates Named Entity Recognition (NER) to address the limitations of generic Question Generation strategies. By shifting the focus from broad contextual understanding to the targeted verification of medical entities, we aim to enhance the system’s sensitivity to high-stakes terminology errors. This entity-aware approach is designed to provide a more granular and clinically relevant evaluation, ensuring that the preservation of vital medical information is prioritized over stylistic fluency.

The entity-aware pipeline processes each source sentence through three sequential stages. First, we extract biomedical entities using a specialized NER model. Second, we generate

questions conditioned on each extracted entity. Third, we answer these questions in both the source and backtranslated contexts, comparing the answers to assess translation quality.

Given a source sentence s , we extract a set of biomedical entities $\mathcal{E} = \{e_1, \dots, e_m\}$ using a dedicated NER model [2]. Each entity e_j is represented as a tuple:

$$e_j = (\text{text}_j, \tau_j, p_j, \text{start}_j, \text{end}_j) \quad (6)$$

where text_j is the entity surface form, $\tau_j \in \mathcal{T}$ denotes its semantic type, $p_j \in [0, 1]$ is the prediction confidence, and $(\text{start}_j, \text{end}_j)$ are the character-level span offsets.

For each extracted entity e_j (Fig. 5), we generate a question q_j that specifically targets the information conveyed by that entity in the context of the source sentence. Unlike the baseline approach, which generates questions about the sentence as a whole, entity-conditioned generation constrains the question’s focus to a specific biomedical concept.

The question generation is formulated as a conditional language generation task. Given the sentence s , the entity text text_j , and the entity type τ_j , we prompt Qwen2.5-3B-Instruct [12]:

$$q_j = \text{LLM}(s, \text{text}_j, \tau_j; \theta_{\text{QG}}) \quad (7)$$

where θ_{QG} denotes the parameters of the question generation model.

The prompt instructs the model to generate exactly one clear, specific question such that: (i) the question can be answered using only information present in the sentence, (ii) the question specifically asks about the designated entity, and (iii) the expected answer is the entity itself or information directly related to it. The generation uses a low temperature ($T = 0.3$) to encourage deterministic, focused questions.

D. Prompting Strategies

To systematically evaluate the impact of prompting strategies on model performance, we implement three distinct prompt variants, each designed to address specific aspects of response generation and quality assessment.

- **P1** (Few-Shot Domain Expert) [1]: Two curated Q&A examples that demonstrate response structure and detail level.
- **P2** (Chain-of-Thought) [11]: Explicit reasoning articulation before the final answer, following recent findings that CoT reduces hallucinations in knowledge-intensive tasks [13].
- **P3** (Concise Output): Responses limited to a maximum of five words, aimed at minimizing noise in string-based similarity computation.

While the baseline evaluation utilizes Few-Shot prompting, our approach incorporates a methodological refinement that warrants explicit delineation. The baseline configuration invokes the model once per sample, generating all responses within a single inference pass. Conversely, our Prompt Strategy executes independent model calls for each question. This architectural distinction yields measurable performance

improvements, especially with the use of a lightweight model. Single-inference approaches exhibit susceptibility to inconsistent response formatting and elevated noise levels when processing multiple questions concurrently. By decoupling each question into discrete inference operations, we ensure uniform attention allocation and structural consistency across responses, thereby mitigating variability in formatting conventions and terminology extraction. The empirical results substantiate both the inherent efficacy of Few-Shot prompting and the incremental benefits derived from granular inference segmentation.

III. EXPERIMENTS

A. Datasets

We evaluate our extensions on two biomedical translation quality datasets introduced by [4].

a) *BioMQM*: Our primary evaluation uses BioMQM, a biomedical machine translation dataset annotated by professional translators using the Multidimensional Quality Metrics (MQM) framework. The dataset contains 5,216 sentence-level instances across five language pairs: English–German (en-de), English–Spanish (en-es), English–French (en-fr), English–Russian (en-ru), and English–Chinese (en-zh). Each instance includes the English source, machine-translated target, back-translation, and MQM severity labels: *Neutral* (no error), *Minor*, *Major*, and *Critical*. This stratification enables fine-grained analysis of pipeline performance across error severities.

Figures 2, 3, and 4 (Appendix B) show baseline performance degrading from Neutral to Critical errors across all metrics.

b) *ContraTICO*: We additionally experiment with ContraTICO, a contrastive dataset of synthetic translation errors in the COVID-19 domain, constructed via controlled linguistic perturbations (e.g., omission, alteration, synonym substitution). While useful for ablation studies, we relegate its analysis to Appendix J and focus on BioMQM, which better reflects naturally occurring errors in real-world biomedical translation.

B. Metrics

For each question q_i , we compare the source and backtranslated answers using five complementary metrics:

- **Token-level F1**: harmonic mean of precision and recall over whitespace-tokenised answers;
- **Exact Match (EM)**: binary indicator of normalised string equality;
- **BLEU** [6]: n -gram precision with brevity penalty;
- **chrF** [7]: character-level F -score;
- **SBERT** [8]: cosine similarity between sentence embeddings.

The sentence-level score under metric ϕ is the average over all k question-answer pairs:

$$\Phi(s) = \frac{1}{k} \sum_{i=1}^k \phi(a_i^{(s)}, a_i^{(bt)}) \quad (8)$$

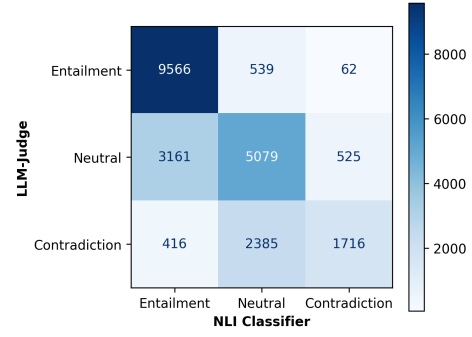


Fig. 1: Confusion Matrix: LLM-Judge vs NLI Classifier

A high score indicates that the translation preserves the semantic content of the source, while a low score signals potential translation errors.

C. NLI Metric extension

Fig. 1 presents the confusion matrix between the two classification approaches across all 23,449 question-answer pairs, showing an overall agreement rate of **69.77%**. We can notice that they agree more when they output entailment, while they often disagree about contradiction.

Table I compares the average probability assigned to each label.

TABLE I: Average Confidence

Label	Confidence (%)	
	LLM	NLI
ENTAILMENT	96.66	90.91
NEUTRAL	89.29	88.07
CONTRADICTION	87.57	81.15

We observe that the LLM is overall more confident about the predictions, but this does not mean that the outputs are better. The prediction of the baseline with our lightweight model are significantly worse than the ones of the original paper. Using SBERT, the decision accuracy is 52.33%, while using F1 it is 48.04%. The main problem that leads to this degradation of performance is discussed in the following extension. Instead, to evaluate the performance of the results obtained with the NLI task, we define a simple function to decide whether the translation is accepted or rejected [4]: if at least a contradiction is present among the pairs of answers to the question we reject it, otherwise we accept it. For the NLI Classifier this leads to surprisingly good results, with a decision accuracy of 75.67%, while the results using the LLM, with an accuracy of 51.69%, are comparable to the ones of the baseline. The low performance of the LLM are probably explainable by the fact that the model we used is very small and general-purpose (see Figures 6 and 7 in Appendix I)

D. NER extension

A critical aspect of quality estimation effectiveness lies in discriminative power—the system’s ability to differentiate

between severity levels through measurable performance gaps. Analysis of consecutive severity transitions (Table II) reveals fundamentally different behavioral patterns between the two approaches.

TABLE II: F1 variation across severity transitions

Severity Transition	$\Delta F1$			
	Baseline	(%)	NER	(%)
Neutral – Minor	−0.0623	−13.89%	−0.0006	−0.08%
Minor – Major	+0.0224	+4.66%	−0.0749	−10.16%
Major – Critical	−0.0503	−10.46%	−0.0680	−9.23%
Neutral – Critical	−0.0902	−18.75%	−0.1435	−19.47%

Despite the similarity of variation in F1 score between Neutral and Critical errors (−18.75% vs −19.47%), the baseline exhibits relatively significant F1 variation across severity levels, except for the Minor-Major transition, where the F1 score of Major severity errors is greater than Minor ones, which explains the bad performance encountered previously. In contrast, the NER-extension demonstrates progressively increasing F1 variation as severity intensifies, with performance gaps that widen substantially as errors become more critical. This progressive widening indicates that the entity-focused approach becomes increasingly selective as error criticality intensifies, capturing the incremental increase in semantic complexity across the severity continuum (see Appendix [sec:ner analysis] for detailed transition analysis). The NER-guided system achieves better-distributed discriminative behavior across consecutive transitions, providing nuanced quality signals that align with the hierarchical nature of translation risk in medical contexts.

E. Prompt-Ablation

Analyzing consecutive severity transitions (Table III) we can compare the baseline with P1, P2 and P3.

TABLE III: F1 variation across severity transitions

Severity Transition	$\Delta F1$ (%)			
	Baseline	P1	P2	P3
Neutral – Minor	−13.89%	−12.31%	−14.73%	−14.96%
Minor – Major	+4.66%	−0.17%	+1.18%	−0.0%
Major – Critical	−10.46%	−9.74%	−11.39%	−4.92%
Neutral – Critical	−18.75%	−22.22%	−24.95%	−19.89%

Notably, while the prompt-enhanced variants largely mitigate the baseline’s anomalous Minor–Major behavior (+4.66%), they do not eliminate it entirely: P1 achieves monotonicity, but just barely (−0.17%), P3 remains unchanged (−0.0%), and P2 retains a small residual instability (+1.18%).

Examining the global performance delta between Neutral and Critical errors, which represents the system’s overall capacity to distinguish low-stakes from high-stakes discrepancies, we can see that the baseline approach exhibits an

F1 variation of −18.75%, while the Few-Shot strategy (P1) demonstrates a larger performance gap of −22.22%. The Chain-of-Thought approach (P2) shows even more pronounced discrimination (−24.95%), though with increased computational overhead, while the Concise variant (P3) exhibits comparable discriminative capacity (−19.89%, only marginally better than the baseline), suggesting that prompt simplification may compromise the system’s ability to capture the hierarchical nature of translation risk in medical contexts. Qualitative analyses comparing all prompting strategies are provided in Appendix H, illustrating the distinct behavioral patterns and error characteristics of each approach.

IV. CONCLUSIONS AND TAKEAWAYS

This work demonstrates that targeted architectural refinements to question-answering pipelines for translation enhance discriminative power and interpretability of monolingual quality estimation in specialized domains.

Our NER-guided extension achieves more gradual score variations between Neutral and Critical errors, indicating enhanced sensitivity to clinically significant translation failures. Domain-specific question generation outperforms generic approaches when errors disproportionately affect critical terminology. Prompt engineering also reveal better discriminative behavior compared to the baseline (which has −18.75% variation between the F1 score of Neutral errors and the score of Critical ones): Few-Shot (P1) demonstrates −22.22% Neutral–Critical delta, while Chain-of-Thought (P2) shows the most pronounced discrimination (−24.95%), at a slightly increased token cost. Replacement of LLaMA-3 70B with Qwen2.5-3B introduces capacity limitations, leading to overall worse prediction accuracy, especially when using the F1 score. This can be partially solved by introducing an NLI task that identifies contradictions in the answers to use as a guide to decide whether to accept or reject the translation. Using a lightweight LLM for this task, instead, leads to no improvements, despite the fact that it is very confident in the answers it gives.

Future work should explore scaling to larger models, extending to additional specialized domains, and investigating active learning strategies where human feedback iteratively improves calibration. These findings contribute to making translation quality estimation more accessible and reliable for monolingual users in high-stakes domains.

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APPENDIX A

PROMPTS

A. LLM-as-Judge

Prompt A.1: LLM-as-Judge

You are a judge evaluating the relationship between two answers to the same question.

Answer A (Source): {answer_src}

Answer B (Backtranslation): {answer_bt}

Classify the relationship as one of:

ENTAILMENT: Answer B supports or is consistent with Answer A.

NEUTRAL: Answer B is neither clearly supportive nor contradictory.

CONTRADICTION: Answer B contradicts or is inconsistent with Answer A.

Respond with **ONLY** the label: ENTAILMENT, NEUTRAL, or CONTRADICTION.

Label:

B. Prompt Ablation

Prompt A.2: P1 (Few-Shot)

Answer medical questions based on the context.

Example 1:

Context: Patient diagnosed with diabetes mellitus type 2, prescribed metformin 500mg twice daily.

Question: What medication was prescribed?

Answer: metformin 500mg

Example 2:

Context: MRI revealed a 2cm lesion in the left frontal lobe.

Question: What was the size of the lesion?

Answer: 2cm

Now answer:

Context: {sentence}

Question: {question}

Prompt A.3: P2 (Chain-of-Thought)

Answer the question about the medical text.

Context: {sentence}

Question: {question}

Think step by step:

1. Identify the key medical information in the context
2. Locate the specific answer to the question
3. Provide the exact answer from the text

Format your response as:

REASONING: [your reasoning]

FINAL ANSWER: [concise answer using exact terms from context]

Prompt A.4: P3 (Concise Constraint)

Context: {sentence}

Question: {question}

Answer in maximum 5 words using exact terms from the context:

C. NER extension

Prompt A.5: Entity-Aware Question Generation

Task: Generate a specific question about the given entity based on the sentence.

The entity is: "{entity}" (Type: {entity_type})

The sentence is: "{sentence}"

Generate exactly ONE clear, specific question that:

1. Can be answered using **ONLY** information from the sentence
2. Specifically asks about the entity mentioned
3. The answer should be the entity itself or information directly related to it

Output only the question, nothing else.

Question:

Prompt A.6: Entity-Aware Question Answering

Task: Answer the question based **ONLY** on the given sentence.

Sentence: {sentence}

Question: {question}

Instructions:

- Answer using **ONLY** information from the sentence
- If the answer is not in the sentence, respond with "[NOT FOUND]"
- Be concise and direct

Answer:

APPENDIX B BASELINE

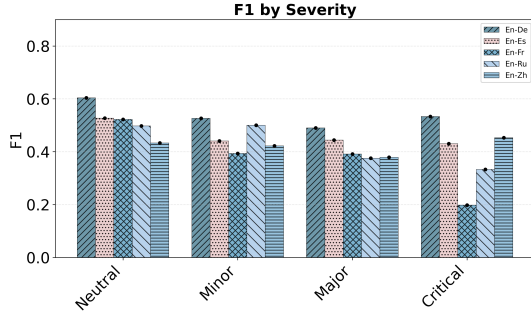


Fig. 2: F1 by severity. Baseline performance across MQM severity levels for all five language pairs showing consistent decrease from Neutral to Critical errors.

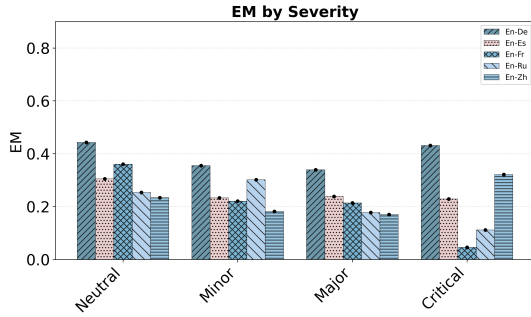


Fig. 3: EM by severity. Exact Match scores across MQM severity levels for all five language pairs showing consistent decrease from Neutral to Critical errors.

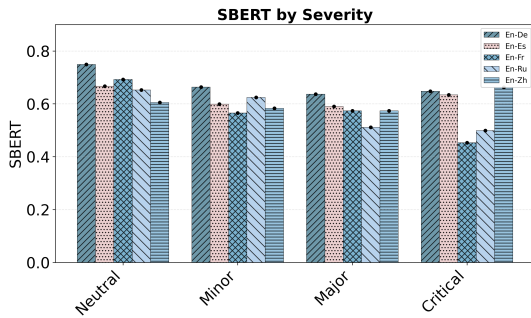


Fig. 4: SBERT by severity. SBERT exhibits greater resilience to severity variations across all five language pairs, although a downward trend remains observable.

APPENDIX C METRICS EXTENSION

In this appendix, we present a breakdown of the agreement rates across different severity levels, showing the number of matches, total items, and the agreement rate for each severity category, as shown in Table IV.

Severity	Matches	Total	Agreement Rate (%)
Neutral	6,757	9,713	69.57
Minor	13,759	20,037	68.67
Major	7,640	10,505	72.73
Critical	451	677	66.62

TABLE IV: Severity Breakdown of Agreement Rates

APPENDIX D NER

In this section, we present a detailed analysis of the performance of the NER-based extension.

Table V shows the performance of NER-based extension.

Severity	F1	EM	chrF	BLEU	SBERT
Neutral	0.737	0.493	75.49	59.43	0.846
Minor	0.737	0.500	75.11	59.69	0.834
Major	0.662	0.469	70.38	53.09	0.821
Critical	0.594	0.419	58.82	45.79	0.682

TABLE V: Performance metrics across severity levels: NER-extension with BioMQM dataset.

Table VI presents the detailed results covering all evaluation metrics divided by language and severity.

Language	Severity	F1	EM	chrF	BLEU
DE	Neutral	0.594	0.434	65.956	53.438
	Minor	0.513	0.347	58.733	44.843
	Major	0.476	0.328	55.037	41.585
	Critical	0.459	0.348	51.662	40.809
ES	Neutral	0.492	0.264	55.151	40.523
	Minor	0.414	0.210	47.631	33.193
	Major	0.423	0.210	47.700	33.649
	Critical	0.410	0.207	48.275	33.563
FR	Neutral	0.483	0.304	57.916	41.254
	Minor	0.352	0.184	41.776	27.853
	Major	0.341	0.186	40.867	27.780
	Critical	0.207	0.060	27.759	13.489
RU	Neutral	0.460	0.240	53.498	36.383
	Minor	0.485	0.294	53.860	40.470
	Major	0.366	0.169	43.068	29.574
	Critical	0.364	0.117	47.090	30.880
ZH-CN	Neutral	0.420	0.219	49.210	33.882
	Minor	0.400	0.160	46.608	29.306
	Major	0.340	0.137	41.184	25.831
	Critical	0.440	0.273	52.403	39.000

TABLE VI: Detailed performance metrics for the NER extension, stratified by language and error severity.

Fig. 5 illustrates how the Exact Match (EM) score is distributed across different entity types found by the NER model.

APPENDIX E PROMPT ABLATION

Table VII shows the results of our prompt ablation study across five languages and four severity levels.

Entity Performance: Exact Match Rate Distribution

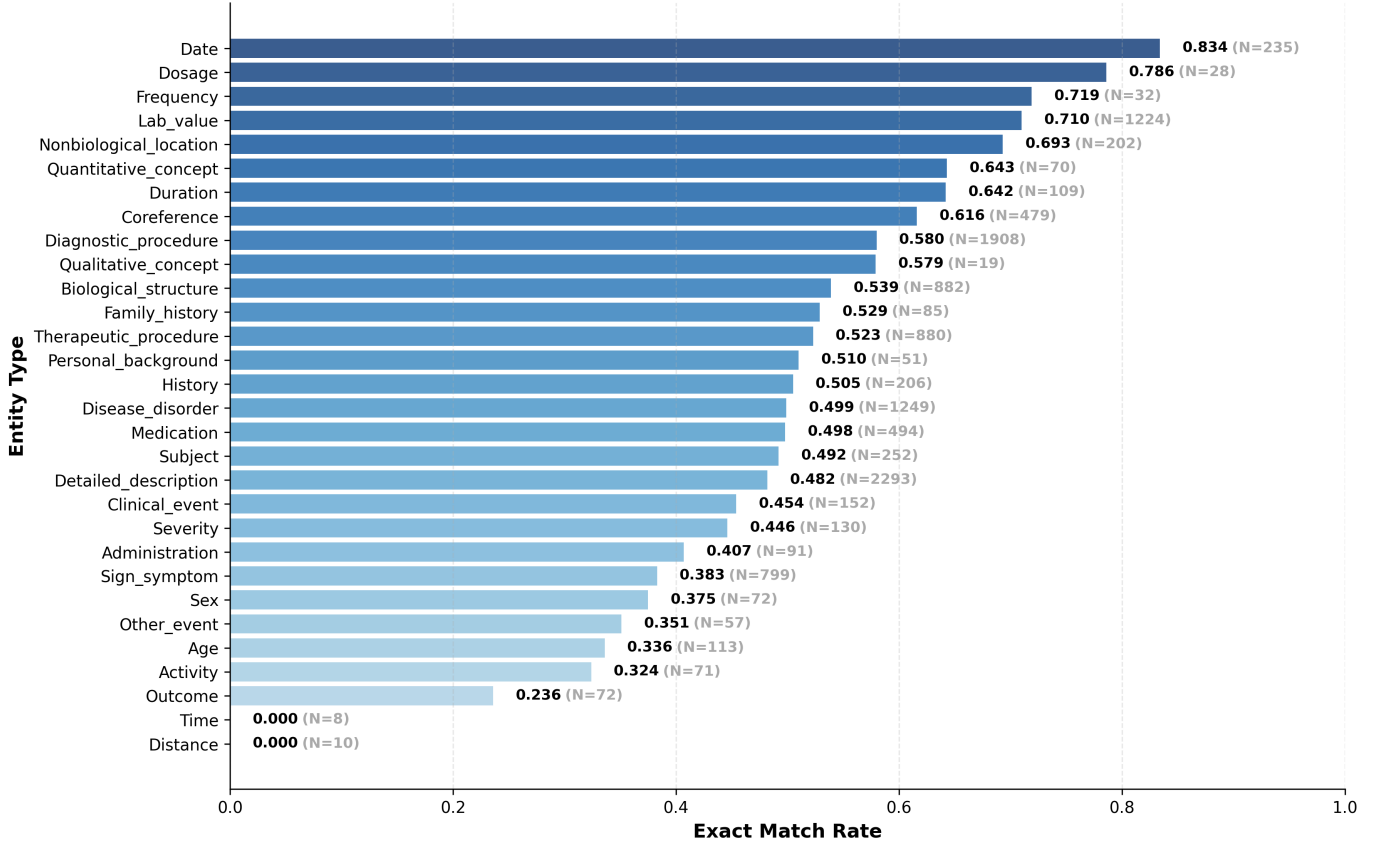


Fig. 5: Distribution of Exact Match (EM) scores across Entity Type.

Lang	Severity	F1	EM	chrF	BLEU	SBERT
DE	Neutral	0.594 / 0.698 / 0.576 / 0.626	0.434 / 0.392 / 0.380 / 0.401	65.956 / 69.892 / 59.175 / 63.820	53.438 / 55.273 / 44.798 / 50.381	0.728 / 0.795 / 0.700 / 0.737
	Minor	0.513 / 0.590 / 0.509 / 0.524	0.347 / 0.281 / 0.337 / 0.305	58.733 / 61.119 / 53.502 / 54.445	44.843 / 42.675 / 38.414 / 40.139	0.650 / 0.716 / 0.637 / 0.642
	Major	0.476 / 0.542 / 0.466 / 0.475	0.328 / 0.250 / 0.285 / 0.265	55.037 / 56.881 / 49.148 / 51.083	41.585 / 38.849 / 33.418 / 35.581	0.619 / 0.690 / 0.606 / 0.620
	Critical	0.459 / 0.441 / 0.414 / 0.381	0.348 / 0.125 / 0.313 / 0.167	51.662 / 45.951 / 47.175 / 40.328	40.809 / 26.016 / 35.800 / 26.432	0.577 / 0.608 / 0.565 / 0.505
ES	Neutral	0.492 / 0.623 / 0.514 / 0.550	0.264 / 0.278 / 0.269 / 0.341	55.151 / 62.511 / 53.493 / 57.764	40.523 / 45.165 / 36.706 / 42.234	0.629 / 0.738 / 0.634 / 0.660
	Minor	0.414 / 0.522 / 0.432 / 0.453	0.210 / 0.193 / 0.214 / 0.228	47.631 / 53.909 / 46.802 / 48.448	33.193 / 35.455 / 30.911 / 32.599	0.570 / 0.667 / 0.572 / 0.587
	Major	0.423 / 0.514 / 0.454 / 0.447	0.210 / 0.176 / 0.241 / 0.224	47.700 / 52.968 / 47.513 / 46.617	33.649 / 33.852 / 33.825 / 31.980	0.565 / 0.670 / 0.593 / 0.581
	Critical	0.410 / 0.467 / 0.403 / 0.431	0.207 / 0.173 / 0.224 / 0.227	48.275 / 51.243 / 44.551 / 46.805	33.563 / 31.426 / 27.983 / 31.415	0.598 / 0.628 / 0.560 / 0.567
FR	Neutral	0.483 / 0.573 / 0.525 / 0.535	0.304 / 0.264 / 0.325 / 0.293	57.916 / 59.845 / 55.774 / 57.064	41.254 / 42.323 / 39.003 / 42.374	0.661 / 0.732 / 0.671 / 0.678
	Minor	0.352 / 0.449 / 0.387 / 0.396	0.184 / 0.156 / 0.196 / 0.195	41.776 / 47.118 / 40.232 / 43.186	27.853 / 29.455 / 25.803 / 29.066	0.532 / 0.611 / 0.549 / 0.566
	Major	0.341 / 0.435 / 0.401 / 0.408	0.186 / 0.135 / 0.247 / 0.220	40.867 / 47.119 / 42.760 / 44.546	27.780 / 28.151 / 30.130 / 31.521	0.539 / 0.634 / 0.569 / 0.575
	Critical	0.207 / 0.452 / 0.280 / 0.346	0.060 / 0.082 / 0.122 / 0.041	27.759 / 50.379 / 35.320 / 42.432	13.489 / 23.471 / 15.703 / 19.323	0.486 / 0.693 / 0.535 / 0.570
RU	Neutral	0.460 / 0.536 / 0.475 / 0.489	0.240 / 0.146 / 0.210 / 0.232	53.498 / 56.201 / 50.756 / 52.488	36.383 / 33.418 / 31.534 / 35.801	0.616 / 0.702 / 0.622 / 0.633
	Minor	0.485 / 0.557 / 0.493 / 0.528	0.294 / 0.198 / 0.264 / 0.282	53.860 / 56.597 / 52.200 / 54.431	40.470 / 37.797 / 36.876 / 39.590	0.613 / 0.707 / 0.626 / 0.643
	Major	0.366 / 0.448 / 0.362 / 0.390	0.169 / 0.069 / 0.169 / 0.171	43.068 / 46.615 / 39.746 / 42.351	29.574 / 23.245 / 24.573 / 27.542	0.500 / 0.606 / 0.491 / 0.508
	Critical	0.364 / 0.398 / 0.329 / 0.444	0.117 / 0.025 / 0.117 / 0.100	47.090 / 43.093 / 35.677 / 40.352	30.880 / 19.859 / 21.115 / 26.311	0.535 / 0.562 / 0.443 / 0.478
ZH-CN	Neutral	0.420 / 0.546 / 0.485 / 0.485	0.219 / 0.204 / 0.251 / 0.239	49.210 / 57.325 / 50.411 / 51.457	33.882 / 36.192 / 32.637 / 34.614	0.596 / 0.713 / 0.639 / 0.639
	Minor	0.400 / 0.489 / 0.413 / 0.419	0.160 / 0.104 / 0.149 / 0.151	46.608 / 51.933 / 43.705 / 45.736	29.306 / 29.646 / 25.433 / 27.599	0.566 / 0.678 / 0.578 / 0.573
	Major	0.340 / 0.422 / 0.355 / 0.362	0.137 / 0.106 / 0.155 / 0.130	41.184 / 46.448 / 38.933 / 40.291	25.831 / 24.857 / 23.808 / 23.861	0.535 / 0.615 / 0.525 / 0.528
	Critical	0.440 / 0.569 / 0.393 / 0.376	0.273 / 0.071 / 0.214 / 0.071	52.403 / 61.817 / 51.168 / 48.225	39.000 / 34.959 / 30.001 / 21.251	0.691 / 0.719 / 0.629 / 0.584

TABLE VII: Evaluation metrics by language and severity level. Each cell presents scores for Baseline / P1 / P2 / P3.

APPENDIX F MODELS

Table VIII lists the pre-trained models integrated into our experimental pipeline along with their respective HuggingFace identifiers.

Model	HuggingFace Model Name
QWEN-2.5 3B	Qwen/Qwen2.5-3B-Instruct
SBERT MINILM	sentence-transformers/all-MiniLM-L6-v2
BIOMEDICAL NER	d4data/biomedical-ner-all
BART MNLI	facebook/bart-large-mnli

TABLE VIII: Models used in the experimental pipeline.

APPENDIX G COMPUTATIONAL & TIME EFFICIENCY

To provide a comprehensive overview of the resource requirements, we report the average computational and time efficiency for the different extensions and tasks proposed in this work. Table IX details the execution times measured on a Kaggle environment equipped with a T4 GPU.

Ext.	Task	Details	GPU	Time
NER	QG	QG+NER	T4	00:30
		Source	T4	00:27
	QA	fr	T4	00:18
		es	T4	00:29
		de	T4	00:30
		ru	T4	00:40
		zh-CN	T4	01:00
Prompt	QA	Source,fr,de,ru	T4	04:00
		es	T4	05:00
		zh-CN	T4	09:00
Metrics	Eval.	NLI+LLM	T4	03:11

TABLE IX: Computation Time.

APPENDIX H QUALITATIVE ASSESSMENT

In this section, we conduct a qualitative analysis to evaluate the interpretability and granularity of the proposed method. Table X presents a selection of representative examples from the test set, categorizing them by the prompting strategy employed (P1, P2, P3) and the severity of the translation error (from *Neutral* to *Critical*).

For each entry, we display the source sentence (X_{src}), the target translation (Y_{tgt}), and the generated question (Q_{src}). Crucially, we compare the answer derived from the source text (A_{src}) with the answer obtained from the back-translation (A_{bt}). This side-by-side comparison illustrates the model’s ability to isolate semantic discrepancies and demonstrates how different prompting constraints influence the precision and verbosity of the generated feedback.

APPENDIX I GAUSSIAN MIXTURE MODEL

To determine optimal rejection thresholds for our classification system, we employ a Gaussian Mixture Model (GMM) approach to compute rejection probabilities. Figure 6 illustrates the GMM rejection probability distribution used to establish

the threshold for rejection decisions based on the F1 score, while Figure 7 shows the corresponding distribution for the SBERT metric.

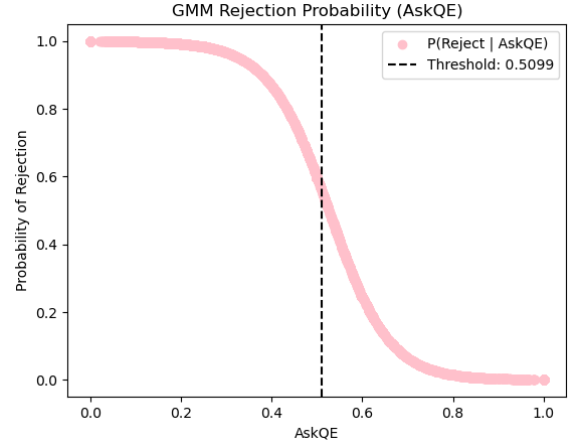


Fig. 6: GMM rejection probability used to compute the threshold of rejection decision for the F1 score.

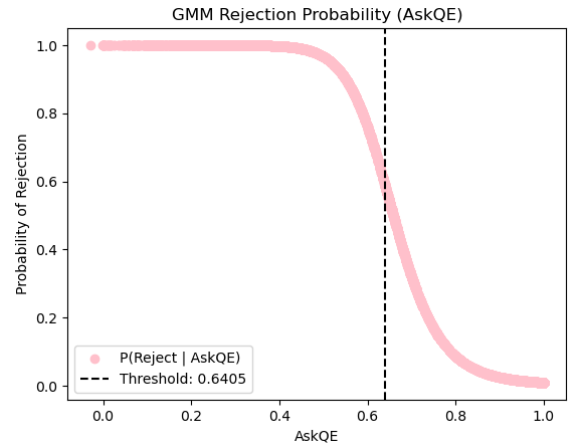


Fig. 7: GMM rejection probability used to compute the threshold of rejection decision for the SBERT metric.

Strategy	Severity	X_{src}	Y_{tgt}	Q_{src}	A_{src}	A_{bt}
P1	Neutral	It can provide high technical and clinical success and low complication rates; it has the potential to substitute the time-consuming balloon-enteroscopy-guided ERCP as well as, in particular, PTCD and, thus, secondary and tertiary therapeutic alternatives.	It can deliver high technical and clinical success and low complication rates; it has the potential to replace time-consuming balloon enteroscopy-guided ERCP and especially PTCD and thus secondary and tertiary therapy alternatives.	What are secondary and tertiary therapeutic alternatives it may offer?	PTCD and secondary therapeutic alternatives.	Secondary and tertiary therapeutic alternatives it may offer include: PTCD
	Minor	Re-interventions became necessary in 6/29 cases.	Reinterventions were necessary in 6/29 cases	How many cases were there initially?	There were initially 29 cases.	Initially, there were 29 cases.
	Major	The features of vertebral pain (VP) and quality of life in postmenopausal women were analyzed depending on the number and location of vertebral fractures (VF).	The characteristics of spinal pain (SP) and quality of life of postmenopausal women were analyzed depending on the number and location of spinal fractures (SF).	What type of pain?	Vertebral pain (VP).	Postmenopausal women
	Critical	There were no established significant differences of quality of life indices according to the EuroQol-5D questionnaire in senior women depending on the number and location of VF, while disturbances of daily activity parameters according to the Roland-Morris questionnaire were found in patients with 2 or more VF.	No significant differences were found in the quality of life scores according to the EuroQol-5D questionnaire in elderly women depending on the number and location of the VN, while impairments in the indicators of daily activities according to the Roland-Morris questionnaire were found in patients with 2 or more VN.	What was measured according to the EuroQol-5D questionnaire?	Quality of life indices	The quality of life scores.
P2	Neutral	Furthermore, the SAX and the ConSept approach were compared regarding differentiation between healthy and diseased myocardium.	In addition, the SAX and ConSept approaches were compared with regard to the differentiation between healthy and diseased myocardium.	Based on what criteria were the approaches compared?	The text implies comparison based on ability to differentiate healthy/diseased myocardium.	Based on the ability to differentiate between healthy and diseased myocardium.
	Minor	Myocardial Mapping in Systemic Sarcoidosis: A Comparison of Two Measurement Approaches.	Myocardial mapping in systemic sarcoidosis: a comparison of two measurement approaches.	What two approaches are being compared in Myocardial Mapping for Systemic Sarcoidosis?	Two measurement approaches are being compared in Myocardial Mapping for Systemic Sarcoidosis.	Two approaches are being compared in Myocardial Mapping for Systemic Sarcoidosis.
	Major	Intravenously administered immunoglobulins and hyperbaric pressure therapy should only be used as supplementary treatments.	Intravenous immunoglobulins and hyperbaric pressure therapy should only be used as adjunctive treatments.	What is the context for using these treatments?	As supplementary treatments	As adjunctive treatments
	Critical	In two pats.	On two clothes.	How many pats?	Two	To properly address this, I need more context or details about what On two clothes refers to. The question How many pats? is not clearly defined within the given context of On two clothes. Could you please provide more information or clarify the full sentence or paragraph?
P3	Neutral	Diagnosis is based on clinical signs such as progressive diffuse swelling, pain, erythema, and early systemic illness.	Diagnosis is based on clinical signs such as progressive diffuse swelling, pain, erythema, and early systemic disease.	What kind of swelling is described?	Progressive diffuse swelling	Progressive diffuse swelling
	Minor	The interdisciplinary surgical cases were identified from the collective and evaluated prospectively with regard to inpatient treatment days, surgical resource expenditure and economic development (DRG before and after microsurgical reconstruction).	The interdisciplinary surgical cases were identified from the collective and prospectively evaluated with regard to inpatient treatment days, surgical resource expenditure and economic development (DRG before and after microsurgical reconstruction).	What type of cases were identified?	Interdisciplinary surgical cases.	Interdisciplinary surgical cases.
	Major	All consecutive pats. with cholangiolithiasis and surgically altered anatomy of the upper GI tract with and without previous attempts of an ERCP were enrolled in this unicenter case study and were characterized with regard to the technical and clinical success of this approach.	All consecutive patients with cholangiolithiasis and surgically altered anatomy of the upper GI tract with and without previous attempts at ERCP were included in this single-center case study and characterized with regard to the technical and clinical success of this approach.	What were the criteria for technical and clinical success?	Technical: No procedural complications. Clinical: No symptomatic failure.	Surgically altered anatomy, ERCP attempts failed.
	Critical	Post-endoscopic, spontaneous perforations and surgical fistulas are included.	The invention also relates to post-endoscopic spontaneous perforations and surgical fistulas.	What types of perforations are included post-endoscopic?	Spontaneous & Surgical perforations.	Post-endoscopic spontaneous perforations, fistulas.

TABLE X: Qualitative Examples of Question and Answer Pairs Generated by ASKQE

APPENDIX J CONTRATICO

A. Metrics Extension

Confusion Matrix: LLM-Judge vs NLI Classifier

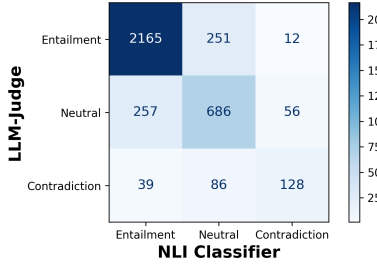


Fig. 8: Confusion matrix atomic

Confusion Matrix: LLM-Judge vs NLI Classifier

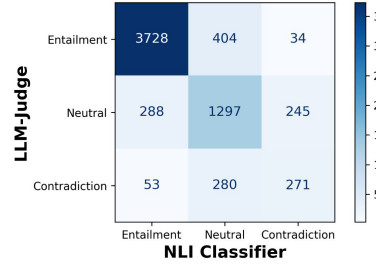


Fig. 9: Confusion matrix semantic

Confusion Matrix: LLM-Judge vs NLI Classifier

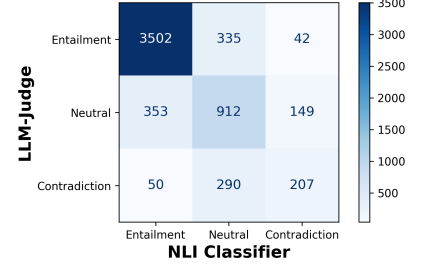


Fig. 10: Confusion matrix vanilla

B. Prompt Ablation

Language	Severity	Perturbation	F1	EM	chrF	BLEU	SBERT
EN-ES	Minor	Spelling	0.658/0.664/0.472/0.474	0.420/0.179/0.206/0.112	73.25/67.47/49.66/49.19	56.55/43.95/27.15/28.19	0.843/0.820/0.635/0.634
		Word Order	0.604/0.659/0.485/0.487	0.356/0.196/0.200/0.130	70.99/66.75/50.00/49.00	50.15/43.27/26.95/27.87	0.812/0.819/0.636/0.642
		Synonym	0.613/0.664/0.478/0.445	0.394/0.191/0.204/0.119	68.31/67.84/50.29/46.55	52.54/44.84/28.59/26.88	0.796/0.830/0.639/0.612
		Intensifier	0.594/0.651/0.465/0.448	0.368/0.169/0.194/0.093	67.84/66.53/50.31/45.66	49.94/42.11/27.52/24.67	0.782/0.822/0.630/0.611
		Expansion (No Impact)	0.601/0.653/0.447/0.442	0.345/0.177/0.173/0.090	69.81/66.14/48.15/46.10	48.78/41.37/24.84/25.04	0.804/0.811/0.608/0.623
	Critical	Expansion (Impact)	0.578/0.641/0.462/0.413	0.311/0.175/0.188/0.092	65.31/65.70/49.10/44.13	45.19/40.53/25.42/23.75	0.772/0.811/0.625/0.606
		Omission	0.547/0.632/0.441/0.429	0.298/0.173/0.142/0.080	63.06/64.63/46.23/44.19	45.35/40.63/23.50/23.86	0.765/0.808/0.618/0.595
		Alteration	0.507/0.573/0.399/0.380	0.259/0.096/0.160/0.084	58.73/60.85/42.74/41.44	40.55/34.27/21.95/21.44	0.722/0.766/0.577/0.572
EN-FR	Minor	Spelling	0.576/0.669/0.492/0.459	0.319/0.204/0.197/0.114	68.13/67.30/51.71/47.34	47.72/44.81/30.43/27.23	0.798/0.816/0.681/0.652
		Word Order	0.573/0.647/0.495/0.455	0.321/0.184/0.193/0.119	64.82/65.15/52.35/46.79	46.52/41.15/29.75/26.64	0.783/0.818/0.688/0.642
		Synonym	0.538/0.656/0.472/0.448	0.281/0.204/0.188/0.113	63.77/65.74/50.65/45.80	43.31/43.92/29.01/25.95	0.781/0.814/0.673/0.638
		Intensifier	0.565/0.678/0.463/0.449	0.336/0.206/0.183/0.082	65.41/68.31/50.78/46.18	46.67/45.39/28.13/25.46	0.778/0.828/0.667/0.645
		Expansion (No Impact)	0.590/0.666/0.483/0.461	0.338/0.186/0.178/0.103	67.28/67.03/51.33/46.77	48.04/44.34/28.39/25.60	0.788/0.820/0.672/0.653
	Critical	Expansion (Impact)	0.551/0.616/0.443/0.420	0.263/0.143/0.154/0.070	62.40/62.94/47.65/43.61	41.96/37.23/24.61/22.75	0.777/0.803/0.643/0.636
		Omission	0.592/0.625/0.471/0.428	0.370/0.160/0.180/0.095	65.37/62.03/49.22/44.38	49.86/39.75/28.92/23.88	0.804/0.784/0.661/0.622
		Alteration	0.459/0.586/0.339/0.339	0.182/0.106/0.117/0.047	54.66/60.91/37.94/37.10	34.47/34.33/18.03/18.48	0.718/0.759/0.549/0.535
EN-HI	Minor	Spelling	0.687/0.606/0.424/0.415	0.451/0.156/0.198/0.112	76.07/60.26/41.91/42.03	61.15/38.34/25.25/24.01	0.857/0.766/0.570/0.591
		Word Order	0.620/0.605/0.399/0.414	0.373/0.146/0.127/0.088	69.90/60.51/39.85/40.04	52.27/36.69/20.69/22.60	0.802/0.772/0.558/0.571
		Synonym	0.593/0.605/0.397/0.442	0.318/0.153/0.163/0.116	68.55/60.36/40.26/43.91	50.07/37.54/23.86/25.90	0.778/0.765/0.549/0.602
		Intensifier	0.666/0.607/0.428/0.409	0.397/0.149/0.189/0.106	74.96/60.31/43.69/41.36	56.30/38.21/25.03/24.39	0.848/0.765/0.589/0.589
		Expansion (No Impact)	0.679/0.607/0.436/0.423	0.424/0.155/0.191/0.100	74.03/61.08/43.37/41.46	58.88/38.84/25.86/23.93	0.832/0.772/0.580/0.585
	Critical	Expansion (Impact)	0.623/0.608/0.417/0.392	0.336/0.147/0.179/0.101	68.61/62.22/42.40/39.83	50.85/38.12/23.77/22.96	0.796/0.772/0.574/0.567
		Omission	0.488/0.573/0.361/0.358	0.273/0.141/0.121/0.052	58.17/56.81/36.27/36.66	40.66/35.07/20.41/18.99	0.725/0.744/0.514/0.532
		Alteration	0.601/0.576/0.386/0.395	0.331/0.136/0.141/0.085	66.17/58.72/40.33/39.70	50.58/35.50/22.05/20.00	0.772/0.751/0.560/0.566
EN-TL	Minor	Spelling	0.666/0.593/0.387/0.351	0.430/0.154/0.159/0.092	72.08/61.06/43.43/38.25	58.52/37.32/24.66/21.96	0.823/0.743/0.567/0.498
		Word Order	0.683/0.588/0.423/0.373	0.443/0.160/0.220/0.100	73.84/60.15/44.11/38.74	59.21/36.81/27.82/22.90	0.834/0.745/0.583/0.508
		Synonym	0.670/0.607/0.376/0.359	0.455/0.158/0.157/0.086	72.45/60.88/40.92/38.01	58.63/38.18/23.18/22.08	0.820/0.752/0.542/0.498
		Intensifier	0.688/0.575/0.417/0.384	0.451/0.165/0.194/0.101	73.08/59.34/44.45/39.75	59.17/35.92/26.15/22.61	0.821/0.744/0.579/0.511
		Expansion (No Impact)	0.657/0.584/0.426/0.384	0.392/0.168/0.193/0.076	73.20/60.86/45.22/40.10	55.23/37.74/27.29/22.21	0.805/0.747/0.586/0.520
	Critical	Expansion (Impact)	0.640/0.583/0.416/0.374	0.357/0.152/0.208/0.079	70.25/60.12/43.85/37.78	52.52/36.75/26.95/20.86	0.798/0.749/0.567/0.513
		Omission	0.600/0.560/0.359/0.345	0.333/0.127/0.157/0.100	66.42/56.46/40.02/37.59	49.61/33.67/22.16/21.84	0.783/0.720/0.550/0.489
		Alteration	0.561/0.555/0.361/0.340	0.292/0.129/0.132/0.057	62.27/57.45/39.34/36.55	46.41/34.62/22.20/20.34	0.752/0.716/0.545/0.493
EN-ZH	Minor	Spelling	0.592/0.657/0.446/0.474	0.311/0.193/0.163/0.106	66.81/65.92/45.57/46.54	48.89/44.29/26.65/26.73	0.782/0.807/0.600/0.657
		Word Order	0.565/0.655/0.488/0.505	0.296/0.206/0.204/0.151	62.90/65.58/49.28/49.70	45.79/43.50/29.74/28.75	0.767/0.812/0.635/0.682
		Synonym	0.555/0.658/0.459/0.494	0.296/0.202/0.188/0.113	64.36/65.56/45.62/47.39	45.77/45.02/26.80/27.24	0.772/0.812/0.620/0.679
		Intensifier	0.540/0.663/0.448/0.501	0.276/0.189/0.185/0.114	63.88/65.55/45.83/48.58	43.86/43.57/27.18/28.16	0.766/0.808/0.604/0.690
		Expansion (No Impact)	0.578/0.638/0.464/0.483	0.331/0.183/0.190/0.108	66.15/63.74/45.99/46.83	47.69/41.42/28.20/27.03	0.771/0.799/0.605/0.674
	Critical	Expansion (Impact)	0.526/0.628/0.441/0.462	0.276/0.171/0.181/0.091	61.35/63.00/44.97/45.71	42.25/40.17/25.68/24.85	0.742/0.787/0.596/0.655
		Omission	0.513/0.603/0.384/0.426	0.256/0.117/0.149/0.091	59.16/59.38/38.17/43.57	41.45/36.34/22.16/24.12	0.735/0.772/0.529/0.630
		Alteration	0.534/0.608/0.396/0.461	0.235/0.128/0.140/0.089	61.90/61.56/40.33/46.51	41.96/37.18/23.62/25.88	0.756/0.769/0.540/0.621

TABLE XI: Performance metrics across language pairs, severity levels, and perturbation types. Each cell shows: Base-line/P1/P2/P3.